156.762 Return on Marketing Investment

Assignment 2

Analysis of the impact of advertising expenditure on The Warehouse sales:

AdStock model research

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1. Title

Analysis of the impact of advertising expenditure on The Warehouse sales: AdStock model research

2. Executive Summary

The purpose of this report is to analyze the impact of spending on different advertising media on retail sales and to evaluate the short- and long-term effects of advertising. We used an actual data set provided by The Warehouse, covering sales data from 20 stores between January 1, 2017, and February 29, 2020. To better capture the lag effect of advertising, we constructed an AdStock model and optimized the explanatory power of the model through logarithmic transformation and adjustment of the lag parameter (lambda).

Through analysis, we found that digital advertising has a significant long-term cumulative effect, with a long-term elasticity reaching 2.7535, while the lagged effect of television and radio advertising, although weaker, still has a positive impact on sales in the long term. In addition, we introduced holiday dummy variables and the interaction term between advertising and weather to control the interference of external factors on sales and ensure the accuracy of the model results.

The results of this report provide advertisers with practical suggestions for optimization of advertising strategies, especially in terms of cross-media allocation of advertising budgets and long-term performance evaluation. Advertisers should focus on the long-term impact of digital advertising while leveraging cross-media synergies to maximize advertising return on investment.

3. Introduction

Advertising plays a vital role in the retail industry. Through effective advertising strategies, companies can significantly increase sales and market share (Smith & Yang, 2015). However, the influence of different advertising media is different. The effect of

advertising is not immediate and usually extends to multiple time periods in the future(Tellis, 2009). This lag effect makes it particularly important to evaluate the short- and long-term effects of advertising. Understanding and quantifying the lag effects of advertising is critical to optimizing ad budget allocation and improving advertising return on investment (ROI).

This report is based on an actual data set provided by The Warehouse, covering sales data and advertising expenditure data for 20 stores from January 1, 2017 to February 29, 2020. In order to deeply explore the lag effect of advertising, we used the AdStock model to capture the cumulative impact of advertising on sales in different time periods by setting the lambda value (0.9). The independent variables of the model include television, radio, digital, mail and newspaper advertising expenditures, and the dependent variable is store sales.

In addition, we introduce holiday dummy variables to control for sales fluctuations during the Christmas and New Year periods. At the same time, the impact of external factors on advertising effectiveness is further controlled through the interaction terms between advertising media and the interaction terms between advertising and weather conditions. The main purpose of this study is to evaluate the impact of different advertising media on sales in the short and long term, quantify their elasticity, and provide advertisers with practical reference when formulating advertising strategies.

Through this analysis, we will reveal the effectiveness of various types of advertising and its different impact on sales in the short and long term, and make recommendations for optimizing the allocation of advertising expenditures to help companies maximize the return on advertising investment.

4. Data Description and Exploratory Data Analysis (EDA)

4.1 Dataset Overview

The dataset used in this analysis is provided by The Warehouse, which contains sales data and advertising expenditure data of 20 stores from January 1, 2017 to February 29, 2020. A total of 1,155 records are included, involving expenditures on multiple advertising media and sales of each store. These data provide rich information support for studying the impact of advertising on sales.

The main features of the dataset (see Appendix Figure 4.1):

Number of observations: 1155

Number of stores: 20

Time range: January 1, 2017 to February 29, 2020

4.2 Data import and variable labels

After importing the data (see code 'Importing Data'), I added descriptive labels to the variables in the dataset to improve the readability of the variables and the accuracy of the analysis. For example, the label of Television is 'Television advertising media spend', and the label of TotalSales is 'Total sales dollars for the above 20 stores indexed to the first day in the data set'. Meteorological variables such as NationalAveAirTemp reflect the average temperature across the country. (see Appendix Figure 4.1)

4.3 Descriptive Statistics

After descriptive statistical analysis of the data, statistical information such as means and standard deviations of various types of advertising expenditures and sales were calculated (see Appendix Figure 4.2). For example, the average expenditure of Television advertising is 1673.80 and the standard deviation is 1007.44; while the average expenditure of Digital advertising is the highest, reaching 2135.34. At the same time, the mean value of total sales is 1.39, with significant fluctuations at different points in time and between stores. Advertising expenditure variables with larger standard deviations indicate higher volatility, which is also reflected in the subsequent graphs.

4.4 Data Exploration and Visualization

To observe time trends and potential outliers in the data, I generated visual charts of key variables (see Appendix Figure 4.3 and 4.4). The line chart shows how various types of ad spend have changed over time, with significant increases evident during holiday periods such as Christmas and New Year's Day. The scatter chart shows the changes in total sales from 2017 to 2020, with higher peak sales during the holidays.

Through these visualizations, I identified several outliers, mostly around the holidays. To handle these outliers, I created holiday dummy variables (see Appendix Figure 4.5) to mark the data between New Year's Day and Christmas. In addition, I adopted a strategy of deleting extreme values, eliminating sales data that were lower than 0.5 or higher than 4.5 during non-holiday periods (based on the quartile method). Removed outliers through conditional statements and created a cleaned data set TWL_DATA_CLEAN_DUMMY (see code 'Data Processing')

5. Model specification

After data cleaning and variable transformation, I built multiple regression models to analyze the impact of advertising expenditures on sales. First, I performed log transformation on advertising expenditures and sales to reduce the impact of skewness and ensure the linear relationship of the model. After that, I performed the following steps:

5.1 Baseline Log-Linear Regression (see Appendix Figure 5.1)

A baseline regression model was constructed using the log-transformed LN_TotalSales as the dependent variable and LN_Television, LN_Radio, LN_Digital, LN_Mailer, and LN_Press as independent variables. The purpose of this model is to measure the linear relationship between advertising expenditure and sales.

5.2 AdStock Variables Calculation (see code 'Calculate Adstock Variables')

The AdStock effect is calculated using a set lambda value (0.9). The AdStock model is

designed to capture the lagged effect of advertising on sales and better reflect the impact of advertising on sales by accumulating advertising spending(Broadbent, 1984). After the calculation is completed, a dataset containing television_adstock, radio_adstock, digital_adstock, mailer_adstock, and press_adstock is generated.

5.3 AdStock Regression (see Appendix Figure 5.3)

Based on the calculated AdStock variable, a regression analysis was performed on LN_TotalSales to measure the impact of the lagged effect of each media advertisement on sales.

5.4 Holiday Dummy Variables Regression (see Appendix Figure 5.4)

To control for the effect of holidays on sales, three dummy variables Holiday_Dummy_2018, Holiday_Dummy_2019, and Holiday_Dummy_2020 are added to mark the holiday effects during New Year's Day and Christmas. This model separates the impact of advertising from holiday sales peaks.

5.5 Interaction Terms Regression (see Appendix Figure 5.5)

Finally, based on the AdStock regression model, we added interaction terms between advertising channels (such as Television * Digital and Radio * Press), as well as interaction terms between advertising and weather conditions (such as Television * NationalAveRainfall). This model aims to capture the interaction between advertising effects and the moderating effect of external conditions on advertising effects.

5.6 Macro for Adstcok(see Appendix Figure 5.6 5.7 and code 'Macro code for adstock')

To further optimize the performance of the AdStock model, I wrote a macro that iterates through different lambda combinations to find the model that maximizes R-squared. In this way, I got the best lambda combination (9_9_9_9_9), and its corresponding adjusted R-squared is 0.3148. This lambda combination corresponds to an AdStock decay coefficient of 0.9 for each advertising medium, which means

that the effect of each advertisement will continue to affect future sales at a rate of 90%.

6. Results

The following conclusions can be drawn from the model results:

6.1 Baseline Log-Linear Regression (see Appendix Figure 5.1)

In the baseline model, the parameter estimates for LN_Television and LN_Digital are 0.00819 and 0.2207 respectively, indicating that these advertising expenditures have a significant positive impact on sales (p-values are both less than 0.05). However, the R-squared value of the model is 0.2441, and the explanatory power is low, indicating that other factors not included still have a greater impact on sales.

6.2 AdStock Regression (see Appendix Figure 5.3)

When the AdStock variable is added, the R-squared value increases to 0.3178, and the explanatory power of the model is enhanced. The coefficients of digital_adstock and television_adstock are significant, indicating that the lagged effect of advertising has a significant positive impact on sales. In particular, the coefficient of digital_adstock is 0.2753, showing that the lag effect of digital advertising is the most significant.

6.3 Holiday Dummy Variables Regression (see Appendix Figure 5.4)

After adding the holiday dummy variable, the R-squared of the model increased slightly to 0.3197. The results show that the impact of holidays is more complicated. The holiday dummy variable in 2019 is positive (0.07954), while the holiday dummy variables in 2018 and 2020 are negative. This shows that the impact of holidays on sales in different years is inconsistent, which may be related to the market conditions of that year.

6.4 Interaction Terms Regression (see Appendix Figure 5.5)

After adding the interaction term, R-squared further improved to 0.3462, and the

explanatory power was improved. Among the interaction terms, the coefficients of Television * Digital and Television * NationalAveRainfall are significant, indicating that there is a synergistic effect between television advertising and digital advertising, and at the same time, the effect of television advertising is affected by rainfall. These results suggest that synergies between different advertising channels and the influence of external conditions play an important role in explaining sales fluctuations.

6.5 Macro for Adstcok(see Appendix Figure 5.6 5.7 and code ’Macro code for adstock’)

The parameter estimation results of this AdStock model are shown in the figure. The estimated coefficients of each AdStock variable (television_adstock, radio_adstock, digital_adstock, mailer_adstock, and press_adstock) are all positive, and most of them are statistically significant (p-value less than 0.05). This shows that the lag effect of advertising on sales is significant. Digital advertising (digital_adstock) has the most significant lag effect on sales, with a parameter estimate of 0.2753, which is much higher than other advertising media. This suggests that digital advertising has the strongest sustained impact when advertising effects are cumulative.

6.6 Short term and Long-term Elasticities

In the optimized AdStock model (with a lambda value of 0.9 for all media), we calculated the short-term and long-term elasticity for each advertising medium:

- Television advertising: short-term elasticity is 0.0393 and long-term elasticity is 0.3927.
- Radio advertising: The short-term elasticity is 0.0286, and the long-term elasticity is 0.2861.
- Digital advertising: The short-term elasticity is 0.2754 and the long-term elasticity is 2.7535.
- Mailer advertising: The short-term elasticity is 0.0144, and the long-term elasticity is 0.1438.

- Press advertising: The short-term elasticity is 0.1068 and the long-term elasticity is 1.0679.

The analysis shows that digital advertising has a significant long-term cumulative effect, with its long-term elasticity far exceeding that of other media (Lambrecht & Tucker, 2013). This finding suggests that advertisers should allocate more budget to digital advertising to maximize its long-term impact.

7. Discussion

The results of this analysis indicate that the impact of advertising expenditures on sales differs significantly across media. Digital advertising, in particular, not only has a strong short-term impact, but also exhibits extremely significant long-term cumulative effects. This suggests that digital advertising should receive greater attention and investment in advertising budget allocation.

At the same time, the lagged effects of television advertising and radio advertising also indicate that these traditional media still have a positive impact on sales within the duration after the advertisement is launched. Although their short-term elasticity is relatively low, the cumulative effect in the long term is still significant. Therefore, Warehouse should consider combining different media in its advertising strategy to make full use of the advantages of each advertising channel.

We successfully control for potential endogeneity issues in the model by introducing holiday dummy variables and interaction terms. These control variables help us better isolate the impact of external factors (such as holidays and weather) on sales fluctuations, resulting in more precise estimates of advertising effectiveness(Wooldridge, 2015). In particular, the interactive effects between advertising media show that cross-media advertising strategies can bring better sales results, and Warehouse should allocate and optimize advertising budgets based on the synergistic effects of different media.

Overall, this analysis provides Warehouse with important insights into optimizing advertising strategies, allocating advertising budgets, and assessing the long-term impact of advertising performance. Warehouse should pay more attention to the cumulative effect of digital advertising, while leveraging cross-media synergy and the moderating effect of external conditions to achieve a higher return on investment.

8. Future Strategies and Model Applications

Based on the results of this analysis, future advertising strategies should be more targeted, especially to make full use of the long-term cumulative effect of digital advertising and the synergy between advertising media. The following are some feasible future strategy suggestions:

8.1 Optimizing cross-media advertising strategies

The analysis results show that the interactive effect of TV advertising and digital advertising has a significant positive impact on sales. Therefore, future Warehouse strategies should pay more attention to collaborative delivery across media. By integrating the broad coverage of TV advertising with the precise delivery capabilities of digital advertising, Warehouse can achieve higher advertising return rates. For example, research shows that integrating multimedia channels can help increase brand awareness and sales. In practice, Warehouse can combine TV and digital advertising to enhance brand influence.

8.2 Focus on investing in digital advertising and continue to run it

This analysis shows that digital advertising has the strongest long-term cumulative effect, and its long-term elasticity is much higher than other advertising media. This is consistent with many research results that digital advertising has a long-term effect in increasing customer interaction and improving brand loyalty. In the future, Warehouse should focus on investing in digital advertising and ensure continuous advertising to fully utilize its cumulative effect. Continuous advertising can maintain consumers' long-term attention to the brand and enhance brand stickiness.

8.3 Leverage data-driven ad optimization

With the development of data science, Warehouse can optimize advertising effects through data-driven advertising strategies. The combination of AdStock model and marketing automation tools provides a scientific basis for advertising, helping companies adjust advertising budgets at the right time. This method can help companies maximize the advertising effect of each time period while reducing resource waste.

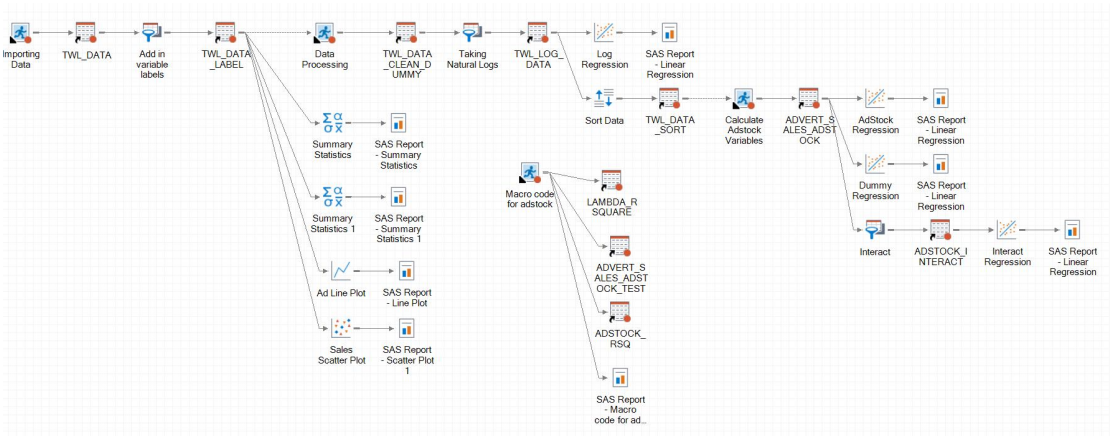
8.4 Dynamic adjustment and testing

The advertising market is a highly dynamic environment, so continuous A/B testing and data monitoring are essential to optimizing advertising results. By testing different combinations of advertising channels and content in real time, Warehouse can quickly adjust advertising strategies to adapt to market changes. This approach not only improves advertising flexibility, but also ensures maximum advertising results through dynamic adjustments.

Through these strategies, companies can make better use of advertising budgets and achieve higher sales growth and return on investment. The application of the AdStock model can not only help companies evaluate the short-term effectiveness of advertising, but also provide important decision-making support for future advertising.

Appendices

[Graph]



SAS work flow1

Summary Statistics

Results

The MEANS Procedure

Variable	Label	Mean	N	N Miss
CalendarDate	Date	21397.00	1155	0
FinYearPlusWeekChar	Character string representing the Financial Year plus the	201877.17	1155	0
Television	Television advertising media spend	1673.80	1155	0
Radio	Radio advertising media spend	724.4552294	1155	0
Digital	Digital advertising media spend	2135.34	1155	0
Mailer	Mailer advertising media spend	970.1420173	1155	0
Press	Press advertising media spend	1151.84	1155	0
NationalAveAirTemp	The national average air temperature across all active weather	12.1112641	1155	0
NationalAveMaxTemp	The national average daily maximum temperature across all	17.9908225	1155	0
NationalAveRainfall	The national average rainfall (mm)	3.3202944	1155	0
Store1Sales	Sales dollars indexed to the first day in the data set	1.4431326	1155	0
Store2Sales	Sales dollars indexed to the first day in the data set	1.4834104	1155	0
Store3Sales	Sales dollars indexed to the first day in the data set	2.0689124	1155	0
Store4Sales	Sales dollars indexed to the first day in the data set	1.3753533	1155	0
Store5Sales	Sales dollars indexed to the first day in the data set	1.2883276	1155	0
Store6Sales	Sales dollars indexed to the first day in the data set	1.4705257	1155	0
Store7Sales	Sales dollars indexed to the first day in the data set	1.2197332	1155	0
Store8Sales	Sales dollars indexed to the first day in the data set	1.4955980	1155	0
Store9Sales	Sales dollars indexed to the first day in the data set	1.3531097	1155	0
Store10Sales	Sales dollars indexed to the first day in the data set	1.2769868	1155	0
Store11Sales	Sales dollars indexed to the first day in the data set	1.1693657	1155	0
Store12Sales	Sales dollars indexed to the first day in the data set	1.3552139	1155	0
Store13Sales	Sales dollars indexed to the first day in the data set	1.1814831	1155	0
Store14Sales	Sales dollars indexed to the first day in the data set	1.5057582	1155	0
Store15Sales	Sales dollars indexed to the first day in the data set	1.1217860	1155	0
Store16Sales	Sales dollars indexed to the first day in the data set	1.1134814	1155	0
Store17Sales	Sales dollars indexed to the first day in the data set	1.6251225	1155	0
Store18Sales	Sales dollars indexed to the first day in the data set	1.9223967	1155	0
Store19Sales	Sales dollars indexed to the first day in the data set	1.6545014	1155	0
Store20Sales	Sales dollars indexed to the first day in the data set	1.4096229	1155	0
TotalSales	Total sales dollars for the above 20 stores indexed to the first day	1.3894698	1155	0

Figure 4.1

Summary Statistics

Results

The MEANS Procedure

Variable	Label	Mean	Std Dev	Minimum	Maximum	N	Lower Quartile	Median	Upper Quartile
Television	Television advertising media spend	1673.80	1007.44	0	5171.61	1155	1012.03	1406.83	2044.82
Radio	Radio advertising media spend	724.4552294	561.4272303	0	2739.38	1155	366.1100000	675.4600000	955.8600000
Digital	Digital advertising media spend	2135.34	1340.34	750.1800000	8189.27	1155	1197.20	1839.00	2554.64
Mailer	Mailer advertising media spend	970.1420173	742.8002999	0	2884.43	1155	360.2200000	958.8600000	1407.93
Press	Press advertising media spend	1151.84	421.6066229	466.4600000	3408.35	1155	887.9900000	1075.79	1370.43
TotalSales	Total sales dollars for the above 20 stores indexed to the first day	1.3894698	0.5503602	0	5.6420000	1155	1.1215000	1.2740000	1.4852000
NationalAveAirTemp	The national average air temperature across all active weather	12.1112641	4.2299102	1.4800000	21.5900000	1155	8.8700000	12.2000000	15.4000000
NationalAveMaxTemp	The national average daily maximum temperature across all	17.9908225	4.3706704	8.1100000	28.1300000	1155	14.2600000	17.6800000	21.7800000
NationalAveRainfall	The national average rainfall (mm)	3.3202944	4.4260963	0	43.0900000	1155	0.4300000	1.6100000	4.6900000

Figure 4.2

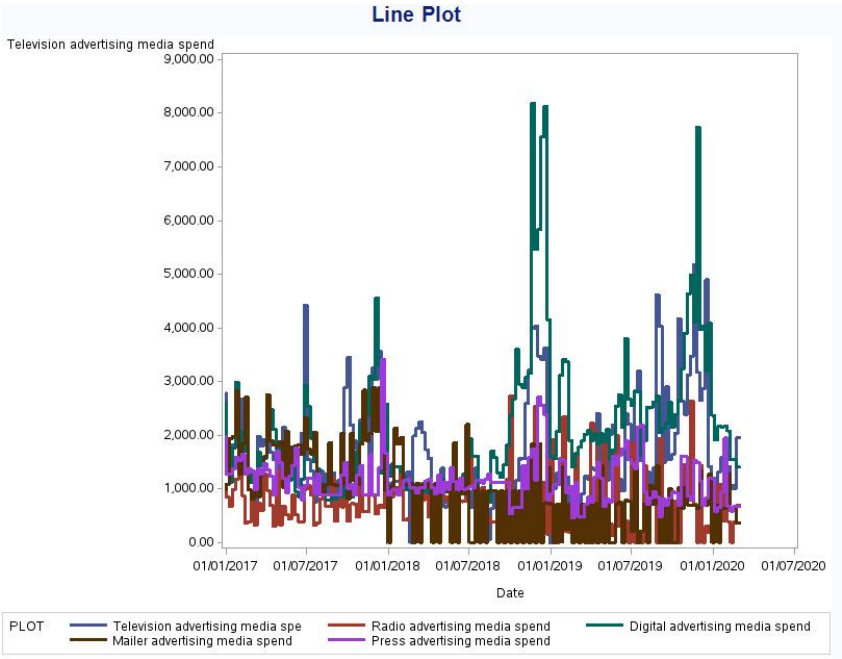


Figure 4.3

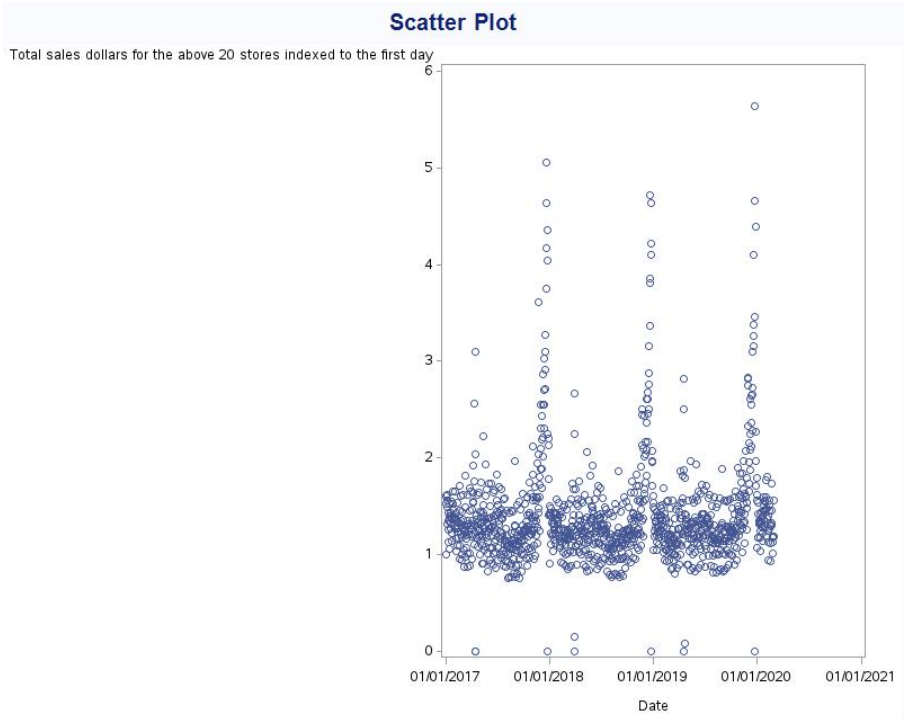


Figure 4.4

Holiday_Dummy_2018	Holiday_Dummy_2019	Holiday_Dummy_2020
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0

Figure 4.5

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_TotalSales LN TotalSales

Number of Observations Read	1140
Number of Observations Used	1140

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	21.15014	4.23003	73.25	<.0001
Error	1134	65.48742	0.05775		
Corrected Total	1139	86.63756			

Root MSE	0.24031	R-Square	0.2441
Dependent Mean	0.28037	Adj R-Sq	0.2408
Coeff Var	85.71124		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	-2.41562	0.16850	-14.34	<.0001	0
LN_Television	LN Television	1	0.00819	0.00666	1.23	0.2187	1.22025
LN_Radio	LN Radio	1	0.00922	0.00331	2.79	0.0054	1.07099
LN_Digital	LN Digital	1	0.22070	0.01536	14.37	<.0001	1.25150
LN_Mailer	LN Mailer	1	0.00706	0.00247	2.86	0.0043	1.06177
LN_Press	LN Press	1	0.12705	0.02213	5.74	<.0001	1.08422

Collinearity Diagnostics								
			Proportion of Variation					
Number	Eigenvalue	Condition Index	Intercept	LN_Television	LN_Radio	LN_Digital	LN_Mailer	LN_Press
1	5.67491	1.00000	0.00005468	0.00066219	0.00344	0.00011572	0.00567	0.00006487
2	0.19163	5.44193	0.00034040	0.00375	0.00661	0.00080031	0.96594	0.00035202
3	0.11221	7.11165	0.00042238	0.01546	0.89655	0.00151	0.00829	0.00044798
4	0.01739	18.06286	0.01256	0.88985	0.05487	0.01014	0.00163	0.01424
5	0.00278	45.17472	0.04448	0.08350	0.03850	0.91863	0.01145	0.18566
6	0.00108	72.43718	0.94214	0.00679	0.00002273	0.06880	0.00701	0.79923

Figure 5.1

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_TotalSales LN TotalSales

Number of Observations Read	1140
Number of Observations Used	1140

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	27.53476	5.50695	105.66	<.0001
Error	1134	59.10279	0.05212		
Corrected Total	1139	86.63756			

Root MSE	0.22830	R-Square	0.3178
Dependent Mean	0.28037	Adj R-Sq	0.3148
Coeff Var	81.42595		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	-3.05759	0.20876	-14.65	<.0001	0
television_adstock		1	0.03927	0.01027	3.82	0.0001	1.50575
radio_adstock		1	0.02861	0.00498	5.74	<.0001	1.23460
digital_adstock		1	0.27535	0.01800	15.29	<.0001	1.54566
mailer_adstock		1	0.01401	0.00448	3.13	0.0018	1.35500
press_adstock		1	0.10679	0.03037	3.52	0.0005	1.19161

Collinearity Diagnostics								
			Proportion of Variation					
Number	Eigenvalue	Condition Index	Intercept	television_adstock	radio_adstock	digital_adstock	mailer_adstock	press_adstock
1	5.86175	1.00000	0.00003032	0.00024213	0.00146	0.00007156	0.00192	0.00002927
2	0.07715	8.71645	0.00089486	0.00544	0.03673	0.00332	0.61504	0.00068142
3	0.05153	10.66552	0.00006010	0.00928	0.77640	0.00082440	0.21249	0.00005181
4	0.00726	28.41770	0.02067	0.79505	0.11349	0.00564	0.06193	0.01852
5	0.00174	57.98687	0.07755	0.18976	0.06662	0.98958	0.08136	0.07726
6	0.00057056	101.35936	0.90080	0.00021957	0.00530	0.00056854	0.02726	0.90345

Figure 5.3

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_TotalSales LN TotalSales

Number of Observations Read	1140
Number of Observations Used	1140

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	27.69565	3.46196	66.43	<.0001
Error	1131	58.94190	0.05211		
Corrected Total	1139	86.63756			

Root MSE	0.22829	R-Square	0.3197
Dependent Mean	0.28037	Adj R-Sq	0.3149
Coeff Var	81.42282		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	-3.08627	0.20994	-14.70	<.0001	0
television_adstock		1	0.03398	0.01123	3.03	0.0025	1.80004
radio_adstock		1	0.02885	0.00498	5.79	<.0001	1.23647
digital_adstock		1	0.28362	0.01899	14.93	<.0001	1.71982
mailer_adstock		1	0.01438	0.00454	3.17	0.0016	1.39363
press_adstock		1	0.10718	0.03050	3.51	0.0005	1.20207
Holiday_Dummy_2018		1	-0.05032	0.07300	-0.69	0.4908	1.01345
Holiday_Dummy_2019		1	-0.08948	0.07983	-1.12	0.2626	1.21219
Holiday_Dummy_2020		1	-0.08824	0.07302	-1.21	0.2271	1.01423

Figure 5.4

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_TotalSales LN TotalSales

Number of Observations Read	1140
Number of Observations Used	1140

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	12	29.99284	2.49940	49.73	<.0001
Error	1127	56.64472	0.05026		
Corrected Total	1139	86.63756			

Root MSE	0.22419	R-Square	0.3462
Dependent Mean	0.28037	Adj R-Sq	0.3392
Coeff Var	79.96190		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	-2.49406	0.23474	-10.62	<.0001	0
television_adstock		1	0.03743	0.01105	3.39	0.0007	1.80685
radio_adstock		1	0.01933	0.00515	3.75	0.0002	1.36867
digital_adstock		1	0.23110	0.02114	10.93	<.0001	2.20857
mailer_adstock		1	0.01723	0.00453	3.80	0.0001	1.43803
press_adstock		1	0.07874	0.03073	2.56	0.0105	1.26483
Holiday_Dummy_2018		1	0.01048	0.07289	0.14	0.8857	1.04778
Holiday_Dummy_2019		1	-0.00145	0.07954	-0.02	0.9855	1.24782
Holiday_Dummy_2020		1	-0.05974	0.07212	-0.83	0.4076	1.02564
Television * Digital	Television * Digital	1	6.520525E-9	4.473765E-9	1.46	0.1453	1.87144
Radio * Press	Radio * Press	1	4.489E-8	2.856995E-8	1.57	0.1164	1.05389
Television * NationalAveRainfall	Television * NationalAveRainfall	1	2.31966E-7	0.00000142	0.16	0.8704	1.03424
Digital * NationalAveAirTemp	Digital * NationalAveAirTemp	1	0.00000780	0.00000157	4.98	<.0001	1.55486

Figure 5.5

Model_Name	Adj_Rsq
AdStock_9_9_9_9_9	0.3148076645

Figure 5.6

The REG Procedure
Model: AdStock_9_9_9_9_9
Dependent Variable: LN_TotalSales LN TotalSales

Number of Observations Read	1140
Number of Observations Used	1140

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	27.53476	5.50695	105.66	<.0001
Error	1134	59.10279	0.05212		
Corrected Total	1139	86.63756			

Root MSE	0.22830	R-Square	0.3178
Dependent Mean	0.28037	Adj R-Sq	0.3148
Coeff Var	81.42595		

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	-3.05759	0.20876	-14.65	<.0001
television_adstock		1	0.03927	0.01027	3.82	0.0001
radio_adstock		1	0.02861	0.00498	5.74	<.0001
digital_adstock		1	0.27535	0.01800	15.29	<.0001
mailer_adstock		1	0.01401	0.00448	3.13	0.0018
press_adstock		1	0.10679	0.03037	3.52	0.0005

Figure 5.7

[Code]

[Importing Data]

```
FILENAME REFFILE 'D:\L Leng\master of analytics\156762 Return on Marketing
Investment\A2 A3\Files for Assignments 2 and
3-20240923/TWL_Data_For_Massey_Protected.xlsx';

PROC IMPORT DATAFILE=REFFILE
    DBMS=XLSX
    OUT=WORK.TWL_DATA
    REPLACE;
    GETNAMES=YES;
RUN;
```

[Data Processing]

```
data work.TWL_DATA_CLEAN_DUMMY;
	set work.TWL_DATA_LABEL;

	/* Delete sales less than 0.5 */
	if TotalSales > 0.5;

	/* Initialize all dummy variables to 0 */
	Holiday_Dummy_2018 = 0;
	Holiday_Dummy_2019 = 0;
	Holiday_Dummy_2020 = 0;

	/* Set dummy for the corresponding peak periods */
	if '01JAN2018'd <= CalendarDate <= '10JAN2018'd then Holiday_Dummy_2018
= 1;
	else if '01JAN2019'd <= CalendarDate <= '10JAN2019'd then
Holiday_Dummy_2019 = 1;
	else if '01JAN2020'd <= CalendarDate <= '10JAN2020'd then
Holiday_Dummy_2020 = 1;

	/* Delete extreme values	greater than 4.5 at other times */
	if (TotalSales <= 4.5) or (Holiday_Dummy_2018 = 1 or Holiday_Dummy_2019 = 1
or Holiday_Dummy_2020 = 1);
run;
```

[Calculate Adstock Variables]

```
/*code to calculate ad stock for advert sales data*/
data work.Advert_Sales_AdStock;
  set work.TWL_DATA_SORT;
```

```
lambda_t = 0.9;
lambda_r = 0.9;
lambda_d = 0.9;
lambda_m = 0.9;
lambda_p = 0.9;

 /* Initialize the adstock variables */
if _N_ = 1 then do;
television_adstock = LN_Television;
radio_adstock = LN_Radio;
digital_adstock = LN_Digital;
mailer_adstock = LN_Mailer;
press_adstock = LN_Press;
retain television_adstock radio_adstock digital_adstock mailer_adstock
press_adstock;
end;

/* Calculate adstock based on lambda values */
else do;
television_adstock = sum(lambda_t * television_adstock, (1 - lambda_t) *
LN_Television);
radio_adstock = sum(lambda_r * radio_adstock, (1 - lambda_r) * LN_Radio);
digital_adstock = sum(lambda_d * digital_adstock, (1 - lambda_d) * LN_Digital);
mailer_adstock = sum(lambda_m * mailer_adstock, (1 - lambda_m) * LN_Mailer);
press_adstock = sum(lambda_p * press_adstock, (1 - lambda_p) * LN_Press);
retain television_adstock radio_adstock digital_adstock mailer_adstock
press_adstock;
end;
```

```
format television_adstock radio_adstock digital_adstock mailer_adstock
press_adstock 8.2;
run;
```

[Macro code for adstock]

```
/* Create dummy dataset to add model and r-squares */
data work.Lambda_Rsquare;
length Model_Name $32. Adj_Rsq 8.;
stop;
run;

%macro Optimise_AdStock_Rsq();
    %do t = 0 %to 9 %by 1;
        %do r = 0 %to 9 %by 1;
            %do d = 0 %to 9 %by 1;
                %do m = 0 %to 9 %by 1;
                    %do p = 0 %to 9 %by 1;

            /* Code to calculate adstock for advert sales data */
            data work.Advert_Sales_AdStock_Test;
                set work.TWL_DATA_SORT;

                /* Calculate lambda for each variable */
                lambda_t = &t.*0.1;
                lambda_r = &r.*0.1;
                lambda_d = &d.*0.1;
                lambda_m = &m.*0.1;
                lambda_p = &p.*0.1;
```

```
/* Initialize the adstock variables */
if _N_ = 1 then do;
    television_adstock = LN_Television;
    radio_adstock = LN_Radio;
    digital_adstock = LN_Digital;
    mailer_adstock = LN_Mailer;
    press_adstock = LN_Press;
    retain television_adstock radio_adstock digital_adstock
mailer_adstock press_adstock;
end;

/* Calculate adstock based on lambda values */
else do;
    television_adstock = sum(lambda_t * television_adstock, (1 -
lambda_t) * LN_Television);
    radio_adstock = sum(lambda_r * radio_adstock, (1 - lambda_r)
* LN_Radio);
    digital_adstock = sum(lambda_d * digital_adstock, (1 -
lambda_d) * LN_Digital);
    mailer_adstock = sum(lambda_m * mailer_adstock, (1 -
lambda_m) * LN_Mailer);
    press_adstock = sum(lambda_p * press_adstock, (1 - lambda_p)
* LN_Press);
    retain television_adstock radio_adstock digital_adstock
mailer_adstock press_adstock;
end;

format television_adstock radio_adstock digital_adstock
mailer_adstock press_adstock 8.2;
run;
```

```
                /* Linear regression for the adstock model */
                proc reg data=work.Advert_Sales_AdStock_Test ADJRSQ
                    outest=work.AdStock_RSq
                    (keep=_model_ _adjrsq_ rename=(_model_=Model_Name
_adjrsq_=Adj_Rsq));
                    AdStock_&t._&r._&d._&m._&p.:
                    Model LN_TotalSales = television_adstock radio_adstock
digital_adstock mailer_adstock press_adstock;
                run;
                quit;

                /* Append the data */
                proc append base=work.Lambda_Rsquare
                    data=work.AdStock_RSq;
                run;

                    %end;
                %end;
            %end;
        %end;
    %end;
%mend;

/* Call the macro to optimize AdStock R-squares */
%Optimise_AdStock_Rsq()
```

References

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